Homework 3

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Model Specification

1. Model formulation (10 points): Write out your model in the form $Y_i = X_i\beta + \epsilon_i$. Describe your mean structure and identify each parameter in β. What does each parameter represent in the context of your research question?

For the mean model, we will take the profile analysis (saturated mean model) approach. We will use reference cell coding. Since this is an observational study, we will not place a constraint on the baseline means and keep baseline as an outcome. This method allows group means (male/female) to differ at baseline.

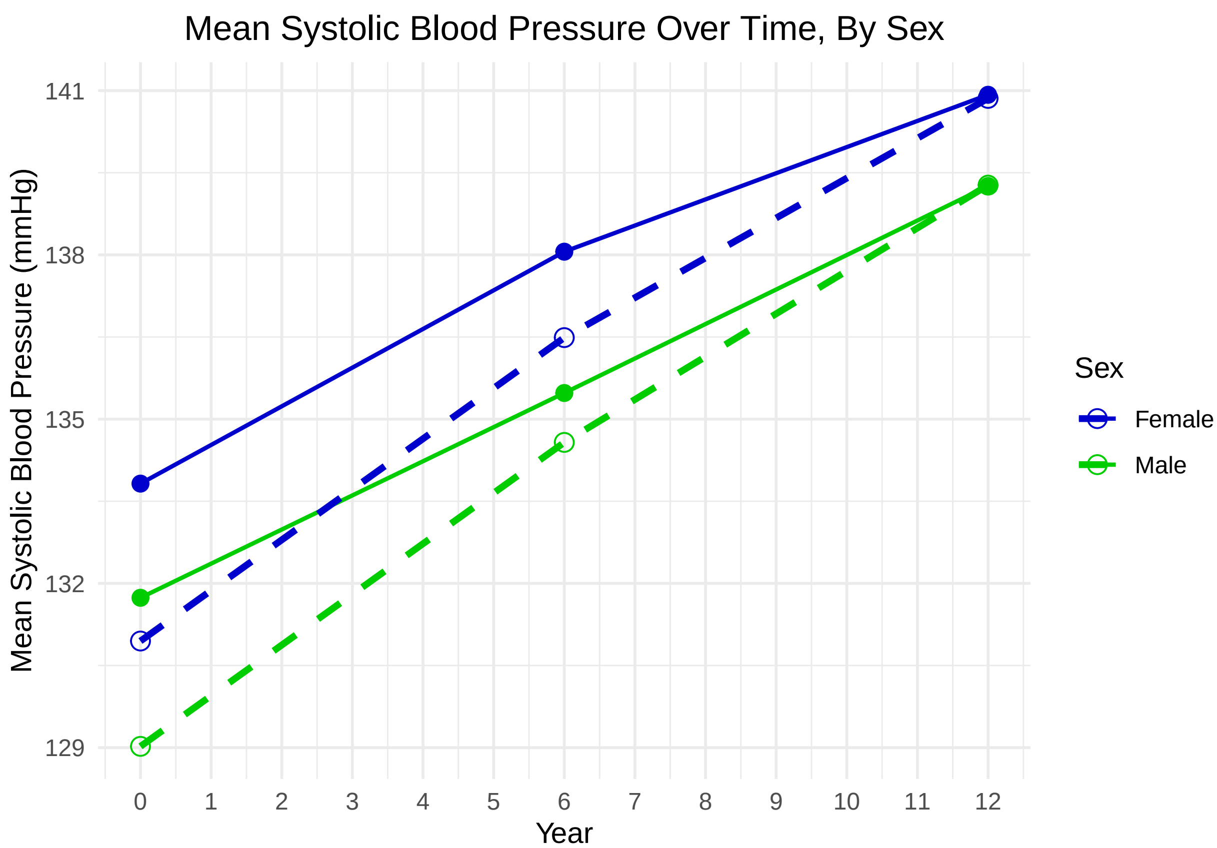
Let $i = 1, ..., 3206$ denote the subjects and let $j = 0, 6, 12$ denote the year of the visit where 0 is the baseline. Recall that patients follow up every 6 years. All patients have a total of 3 visits. We have that $Y_i = (Y_{i0}, Y_{i6}, Y_{i12})^T$, $\beta = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5)^T$. We can write out the following:

$$Y_{ij} = \beta_0 + \beta_1 I(\text{year} = 6) + \beta_2 I(\text{year} = 12) + \beta_3 I(\text{female}) + \beta_4 I(\text{female}) I(\text{year} = 6) + \beta_5 I(\text{female}) I(\text{year} = 12) + \epsilon_{ij}$$

- β_0 denotes the baseline mean SBP for males
- β_1 denotes the change in mean SBP from baseline to year 6 for males
- β_2 denotes the change in mean SBP from baseline to year 12 for males.
- β_3 denotes the difference in mean SBP between females and males at baseline
- β_4 denotes the additional difference in mean SBP between females and males at year 6
- β_5 denotes the additional difference in mean SBP between females and males at year 12
- $E[\epsilon_{ij}] = 0$

2. Visualization (10 points): Create a figure showing your observed data (e.g., individual trajectories or group means over time) along with the fitted mean structure from your model. Does the model appear to capture the patterns in your data?

We fit a WLS/GEE model with working independence and unstructured covariance. The plot below shows the observed data group means as solid lines and the fitted mean structure as dashed lines. The model appears to capture the difference in sex means, but it underestimates the mean SBP for the first two visits.



Estimation and Covariance Structure

1. Covariance structure (15 points): Select a covariance structure for $Var(\epsilon_i)$ and fit your model. Justify your choice based on what you observed in your data (e.g., from the sample correlation matrix or your understanding of the data-generating process).

The following is the estimated correlation matrix for SBP measurements at each visit, constructed using an unstructured covariance structure.

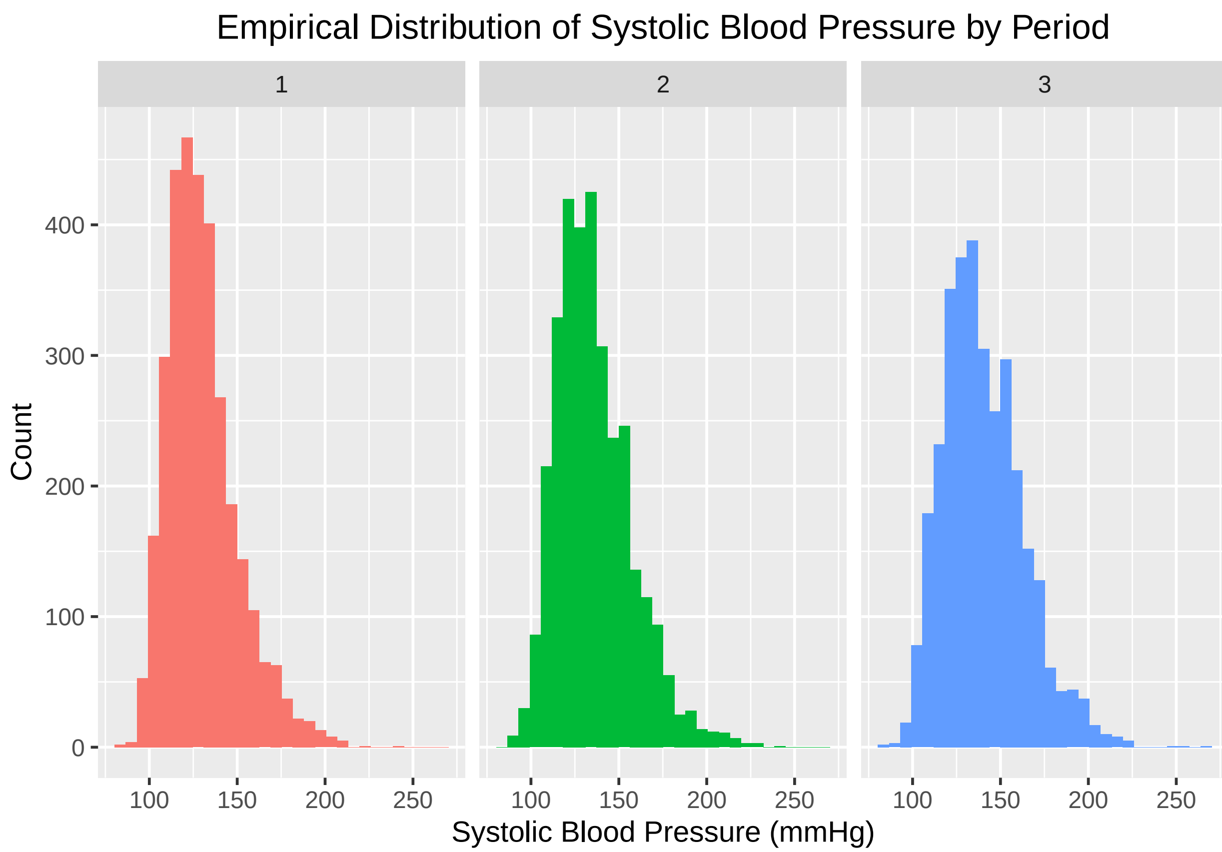
```
## Estimated Correlation Matrix (Unstructured):

##           [,1]      [,2]      [,3]
## [1,] 1.0000000 0.7172575 0.6174050
## [2,] 0.7172575 1.0000000 0.6771183
## [3,] 0.6174050 0.6771183 1.0000000
```

We see that measurements closer in time are more highly correlated (0.717 between baseline and 6 years compared to 0.617 between baseline and 12 years). We also find that SBP at 6 years has a higher correlation with SBP at baseline compared to SBP at 12 years. Based on this, we will choose AR(1) as our covariance structure for $Var(\epsilon_i)$, since we have equally spaced measurements with decaying correlation over time.

2. Estimation method (15 points): Select an estimation method and justify your choice. Report your parameter estimates ($\widehat{\beta}$) with standard errors, and the estimated covariance parameters.

The histograms of systolic blood pressure show that across all periods, the distribution of systolic blood pressure measurements is right-skewed. As such, it is not appropriate to assume that our response variable is normally distributed, so we have been using WLS/GEE as our estimation method, since it does not impose any distributional assumptions on the response variable.

We will fit a WLS/GEE model, assuming an AR(1) covariance structure.



Below we have the estimated covariance paramters, the estimated covariance matrix, and the parameter estimates with standard errors.

```
##
## Estimated covariance parameters:
```

```
##   Parameter Estimate
## 1       rho   0.7435
## 2    sigma2 463.7952
## 3     sigma  21.5359
```

```
##
## Estimated AR(1) covariance matrix:
```

```
##          [,1]     [,2]     [,3]
## [1,] 463.7952 344.8476 256.4059
## [2,] 344.8476 463.7952 344.8476
## [3,] 256.4059 344.8476 463.7952
```

Table 1: GEE Parameter Estimates / w Robust Standard Errors

	Estimate	Robust.SE	Z.statistic	P.value
Intercept	129.0231	0.4589	281.1705	0.0000
Year6	5.5517	0.4135	13.4249	0.0000
Year12	10.2509	0.5091	20.1346	0.0000
Female	1.9237	0.6889	2.7926	0.0052
FemaleYear6	-0.0099	0.5571	-0.0178	0.9858
FemaleYear12	-0.3396	0.6758	-0.5026	0.6153

Algorithm converged after 3 iterations.

Table 2: GEE Standard Errors: Model-Based vs. Robust

	Model.SE	Model.Z	Model.P	Robust.SE	Robust.Z	Robust.P
Intercept	0.5833	221.1833	0.0000	0.4589	281.1705	0.0000
Year6	0.4178	13.2887	0.0000	0.4135	13.4249	0.0000
Year12	0.5516	18.5824	0.0000	0.5091	20.1346	0.0000
Female	0.7694	2.5004	0.0124	0.6889	2.7926	0.0052
FemaleYear6	0.5510	-0.0180	0.9856	0.5571	-0.0178	0.9858
FemaleYear12	0.7276	-0.4668	0.6406	0.6758	-0.5026	0.6153

Variance Estimation

1. Comparison (10 points): Compare the model-based and sandwich (robust) standard errors for your fixed effects. Create a table showing both sets of standard errors

We find that for almost all parameters, the robust standard errors are slightly smaller than the model-based standard errors. Also in general, the both standard errors are very similar. The largest absolute difference is approximately 0.13 between the intercept standard errors.

2. Interpretation (10 points): What do the differences (or similarities) between these standard errors suggest about your covariance model? Given your sample size, which variance estimator do you consider more appropriate for your analysis, and why?

Since most of the standard errors are similar, we can deduce that our working covariance model is reasonable and we can utilize our model based standard errors. Also note that we have a sample size of $n = 3206$, so our sandwich estimators will not have small N bias. The model-based variance estimators are more appropriate for our analysis.

Hypothesis Testing

1. Primary hypothesis (10 points): State and test a hypothesis relevant to your research question using a Wald test. Report the test statistic, degrees of freedom, p-value, and interpret the result.

We will test whether males and females have the same mean SBP at each time point. Since we have a saturated mean model with reference cell coding we want to test if

$$H_0 : \begin{pmatrix} \beta_3 \\ \beta_4 \\ \beta_5 \end{pmatrix} = \mathbf{0}$$

. We can also write this as $Q'\beta = \mathbf{0}$ where $Q' = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$ and $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5)^T$.

$T_W = (\boldsymbol{Q'}\widehat{\boldsymbol{\beta}})'[\boldsymbol{Q'}\widehat{Var}(\widehat{\boldsymbol{\beta}})Q]^{-1}(\boldsymbol{Q'}\widehat{\boldsymbol{\beta}}) \sim \chi^2_3$ under H_0.

After running the test with 3 degrees of freedom, we get a test statistic of 8.49 and a p-value of 0.036. Since the p-value is < 0.05, we reject the null hypothesis and conclude that males and females have different mean SBPs at each time point.

```
##   Statistic DF    P_value
## 1  8.494211  3 0.03682928
```

2. Additional test (10 points): Conduct one additional hypothesis test (e.g., testing a different effect, or testing multiple parameters simultaneously). Report and interpret the results.

We will test whether SBP changes over time. Since we have a saturated mean model with reference cell coding we want to test if

$$H_0 : \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_4 \\ \beta_5 \end{pmatrix} = \mathbf{0}$$

. We can also write this as $Q'\beta = \mathbf{0}$ where $Q' = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$ and $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5)^T$.

$T_W = (\boldsymbol{Q'}\widehat{\boldsymbol{\beta}})'[\boldsymbol{Q'}\widehat{Var}(\widehat{\boldsymbol{\beta}})Q]^{-1}(\boldsymbol{Q'}\widehat{\boldsymbol{\beta}}) \sim \chi^2_4$ under H_0.

After running the test with 4 degrees of freedom, we get a test statistic of 944.26 and a p-value that is less than 0.0001. Since the p-value is < 0.05, we reject the null hypothesis and conclude that SBP changes over time for at least one of the sex groups.

```
##   Statistic DF       P_value
## 1  944.2622  4 4.276045e-203
```

Summary and Interpretation

Summarize your findings as you would for a collaborator. Address:

- What model did you fit and why?
- What are the key findings regarding your research question?
- What assumptions are you making, and are they reasonable?

We fit a GEE model to our data, assuming an AR(1) covariance structure. This assumption is reasonable because the three time periods are equally spaced and, after fitting an initial WLS model with unstructured covariance, we see that the estimated correlation of SBP between different time periods decreases as the time periods get further apart. We estimated that the variance of SBP at each fixed time point is 463.8 and that the correlation between SBP measures at time period t and at time period $t + k$ is 0.744^k for $k \neq t$.

Our hypothesis tests inform us that the population mean trajectories of SBP for woman and for men are different and that for at least one of the sex groups, SBP changes over time. We could further investigate whether or not the two population mean trajectories are parallel, which would indicate that SBP changes at the same rate in both sex groups.